

National University of Computer & Emerging Sciences, Karachi Campus



Research Proposal Of AI-Based Fake News and Misinformation Detector in Pakistan

Nature of Project: Research & Development

Area of Specialization: Mobile App Development

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1. Introduction

The rapid expansion of digital media and social networking platforms has changed the fundamentals of how information is produced, shared, and consumed across the world. In Pakistan, this transformation is much more noticeable. Platforms like WhatsApp, Facebook, Twitter (now known as X), and YouTube serve as primary news sources for the majority of the population. However, this broader access to information has simultaneously made it easier for fake news and misinformation to spread. According to the Digital Rights Foundation, Pakistan ranks among the countries most affected by online misinformation, with fabricated political narratives, health hoaxes, and religiously motivated disinformation circulating at an alarming rate [1].

What makes the Pakistani context challenging is the multilingual nature of online talks. Content spreads not only in formal Urdu or English, but also in Roman Urdu, which is a transliterated hybrid language that blends Urdu vocabulary with the Roman script, which is now widely used in text messaging and social media posts. Existing automated detection tools, which are built only for the English language, are entirely blind to this new language.

Artificial Intelligence, particularly in the domain of Natural Language Processing (NLP), has shown great promise in detecting fake news. Transformer-based models such as BERT, RoBERTa, and multilingual variants like mBERT have achieved close to perfect results on English benchmark datasets. Deep learning architectures, including hybrid CNN-LSTM models, have also shown strong performance. Despite this progress, a significant gap still remains. None of these solutions have been adapted for the Urdu or Roman Urdu languages, and none have been deployed as accessible, real-time mobile applications targeting the Pakistani public.

This project proposes the development of an AI-Based Fake News and Misinformation Detector, a research-driven mobile application that addresses these critical shortcomings. The proposed solution will employ multilingual NLP models fine-tuned on Urdu, Roman Urdu, and English content, integrate real-time fact-checking against verified Pakistani and international news sources, and deliver results through an intuitive, user-friendly Android/iOS mobile application. By bridging the gap between advanced AI research and accessible public tools, this project aims to make a measurable contribution to the fight against misinformation in Pakistan.

2. Related Work

This section synthesizes findings from five research studies and five existing misinformation detection tools, examining their methods, contributions, limitations, and relevance to the proposed idea.

2.1 Review of Research Papers

2.1.1 Hybrid Deep Learning with FastText and Explainable AI (Hashmi et al., 2024)

Hashmi et al. [2] present a comprehensive fake news detection framework that integrates FastText word embeddings with a range of Machine Learning and Deep Learning models, including SVM, Random Forest, LSTM, GRU, BiLSTM, and a hybrid CNN-LSTM architecture. Tested on three publicly available English datasets, WELFake (72,134 articles), FakeNewsNet (23,196 articles), and FakeNewsPrediction, the CNN-LSTM hybrid achieved the highest accuracy of 0.99 across multiple datasets, outperforming transformer-based models such as BERT, XLNet, and RoBERTa. Additionally, the study employs Explainable AI techniques, LIME (Local Interpretable Model-Agnostic Explanations) and LDA (Latent Dirichlet Allocation), to interpret model decisions, identifying keywords that drive classifications. A big limitation acknowledged by the authors is the reliance on only English-language datasets. The authors themselves recommend extending the work to low-resource languages using multilingual transformers such as mBERT and mT5 as future work, a gap that this proposal directly addresses. Furthermore, no mobile application or real-time deployment is provided.

2.1.2 AI and Fake News: A Conceptual Framework (Ameli et al., 2022)

Ameli et al. [3] propose a three-tier conceptual framework for classifying and detecting fake news, structured around characterization and feature extraction, classification and detection, and defence mechanisms. The paper provides a thorough overview of AI's dual role in misinformation, both as a tool for generating fake content and as a means to detect it. The framework classifies fake news sources into traditional media, social media, and search engines, and identifies political context as a separate classification category. However this framework only remains as an idea, and has not been deployed yet. The paper agrees that detection systems need to handle diverse linguistic styles and cultural contexts, yet does not provide any multilingual solution. It also does not address Urdu or Roman Urdu, nor is there any discussion of mobile deployment, leaving a lot of room for practical work.

2.1.3 AI-Based Credibility Indicators Under Social Influence (Lu et al., 2022)

Lu et al. [4] take a behavioural approach to misinformation, conducting two controlled randomized experiments on Amazon Mechanical Turk to study how AI-generated credibility indicators affect users' ability to detect misinformation under social influence. Using COVID-19 news stories and a Naive Bayes classifier with 75% accuracy, the study shows that presenting AI credibility warnings before users form their judgements significantly increases alignment with the AI model's prediction, whether correct or incorrect. A valuable finding is that social influence increases the effect of AI-before credibility indicators. However, the study is fully experimental and does not give a deployable system. The news content is only in English, the model accuracy is intentionally limited, and the setup involves no mobile interface. This paper informs the proposed project's user-interface design, the importance of when and how credibility indicators are presented to users in the mobile app.

2.1.4 Big Data ML-Based Fake News Detection Using Distributed Learning (Altheneyan & Alhadlaq, 2023)

Altheneyan and Alhadlaq [5] address the scalability challenge of fake news detection by proposing a stacked ensemble model deployed on an Apache Spark distributed cluster. Using the FNC-1 dataset (49,972 headline-article pairs with four categories: agree, disagree, discuss, unrelated), the study employs HashingTF-IDF feature extraction with Logistic Regression, Random Forest, and Decision Tree classifiers. The stacked ensemble achieves an F1-score of 92.45%, surpassing the baseline by 9.35%. The study shows that distributed computing frameworks can handle large-scale fake news datasets efficiently. Key limitations are that the dataset is entirely English-language, the four-class categorization does not work well with binary real/fake classification needs, the model does not support real-time or mobile deployment, and no multilingual or regional language support is provided. The authors explicitly recommend extending the work with deep learning models and multilingual support, again, a direction this proposal takes forward.

2.1.5 Explainable Misinformation Detection Across Multiple Social Media Platforms (Joshi et al., 2023)

Joshi et al. [6] propose a combination of Domain Adversarial Neural Network (DANN) and LIME for explainable misinformation detection across multiple social media platforms including Twitter(X), Reddit, YouTube, Instagram. Their CoAID-based experiments show DANN improves F1-score by 40% compared to models without domain adaptation, and increases accuracy by 3% and AUC by 9% on the MiSoVac dataset they selected. The use of DANN allows the model

to generalize from one social media domain to another without requiring labelled target domain data. LIME is then applied to explain predictions, highlighting keywords like "vaccine" and "coronavirus" as decision drivers. Despite these innovations, the work is entirely focused on COVID-19 content in English. No Urdu or Roman Urdu content is considered, and no mobile application is developed. The authors acknowledge that domain adaptation across multiple languages remains an open problem, motivating the multilingual dimension of this proposal.

2.2 Review of Existing Tools

2.2.1 Google Fact Check Explorer

Google Fact Check Explorer (<https://toolbox.google.com/factcheck/explorer>) is a web based tool developed by Google that collects and stores fact-check articles from verified publishers using ClaimReview schema markup. Users can search for any topic and get the fact-check results from credible organizations. Its key strength lies in utilizing a globally distributed network of professional fact-checkers. However, the major weaknesses for this in the context of fake news detection are search results that are mostly in English. Urdu and Roman Urdu content is not supported. Real-time detection is unavailable, and the tool operates reactively, meaning only after a fact-check is published. There is no mobile app, no NLP classifier, and no automated detection. The proposed system would integrate with fact-checking APIs like this one but would add automated multilingual AI detection on top.

2.2.2 Snopes

Snopes (<https://www.snopes.com>) is one of the oldest and well known fact-checking websites, operated by professional human fact-checkers. It covers many topics including politics, health, science, and popular culture. Its decisions (True, False, Mixture, Unproven) are well-respected and carefully researched. However, Snopes is exclusively English-language and relies fully on manual fact-checking, making it inherently slow. There is no automated AI detection, no Urdu or Roman Urdu support, no Pakistani news source integration, and no mobile application. The proposed system would use Snopes' database as one of many reference sources for verification.

2.2.3 ClaimBuster

ClaimBuster (<https://idir.uta.edu/claimbuster>) is an academic tool developed at the University of Texas at Arlington that uses NLP to identify and rank check-worthy factual claims from textual input, mainly on political speech. It uses machine learning models to score how much a given sentence warrants fact-checking. It is very good at automatically detecting claims at large scale. However, ClaimBuster is only trained on English political text and performs poorly on informal text, and languages other than English. It only verifies that the claim should be checked, not that it is true or false. It does not support Roman Urdu or Urdu, has no mobile deployment, and has no Pakistan-specific training data. For the proposed project, ClaimBuster's claim-scoring methodology provides insight for the claim extraction module.

2.2.4 Botometer

Botometer (<https://botometer.osome.iu.edu>) is a tool developed by Indiana University that uses machine learning to predict that a Twitter(X) account is a bot, based on more than a 1,000 features, some of them including tweet content, network structure, and posting patterns. This is highly relevant in the context of misinformation because bot accounts are frequently responsible for generating fake news. However, Botometer's limitation for this project is that it judges accounts, not content. It does not analyze whether specific news is true or false. It operates only on English-language Twitter data, provides no Urdu or Roman Urdu support, and has no mobile app. Botometer's methodology inspires the proposed system's social propagation analysis module, which will flag content that shows bot-like amplification patterns.

2.2.5 Factiveverse

Factiveverse (<https://factiveverse.ai>) is a commercial AI-powered fact-checking API built for media organizations and developers. It uses transformer-based NLP models to verify claims against online sources and provides structured decisions. Compared to the other tools reviewed, Factiveverse is the most technically advanced. It supports multiple languages and integrates with external knowledge bases. However, the languages Urdu and Roman Urdu are yet to be implemented, since its main focus is on Western languages. It is a paid API with no free mobile application, no Pakistan-specific news source integration, and no community-driven or user-input features. The proposed system draws inspiration from Factiveverse's API-based architecture but distinguishes itself through open-source multilingual NLP, Pakistan-centric training data, and a free mobile app targeting Pakistani users directly.

3. Research Gap

The synthesis of the five research studies and five existing tools shows a consistent and critical gap, that despite the significant advances in AI-based fake news detection, no existing research study or commercial tool perfectly addresses misinformation in the Urdu or Roman Urdu languages as encountered on Pakistani social media platforms, and none deliver this detection capability through a real-time, user-accessible mobile application.

Specifically, the gap can be sectioned into three areas. First, there is a language gap. All five reviewed research papers, including those by Hashmi et al. [2], Altheneyan & Alhadlaq [5], and Joshi et al. [6], rely solely on English-language datasets (WELFake, FNC-1, CoAID). All five tools reviewed are also either only English-language (Snopes, ClaimBuster, Botometer) or unverified for Urdu/Roman Urdu support (Google Fact Check Explorer, Factiveverse). Roman Urdu, which is used majorly in informal digital communication in Pakistan across platforms like WhatsApp and Twitter(X), is not addressed at all in the existing literature and tools.

Second, there is a deployment gap. Reviewed studies by Ameli et al. [3] and Lu et al. [4] provide conceptual frameworks and experimental findings but do not produce any deployable system. Hashmi et al. [2] and Joshi et al. [6] train and evaluate models but do not provide a working application interface. All five reviewed tools are web-based platforms requiring desktop access. None of them offer dedicated, offline mobile applications targeting a mass user base in a developing country context where smartphone hackability is high but desktop usage is comparatively low.

Third, there is a Pakistan-specific context gap. Pakistani news sources (Dawn, Geo News, ARY News) are not integrated into any existing detection system. The unique political, cultural, and religious context of misinformation in Pakistan, including narratives around elections, blasphemy, and tensions among sects, is absent from training datasets, resulting in models that are poorly calibrated for Pakistani content even when applied to English-language Pakistani news.

This proposal fills all three areas of this gap by developing a multilingual AI model trained on Urdu, Roman Urdu, and English fake news datasets, integrated with Pakistan-specific news APIs, and delivered as a free mobile application accessible to ordinary Pakistani smartphone users.

4. Purpose of the Study

The purpose of this research and development project is to design, develop, and evaluate an AI-powered multilingual fake news and misinformation detection mobile application specifically calibrated for the Pakistani digital information environment. The application will serve everyday users, students, journalists, and general citizens, by enabling them to verify the credibility of news content in English, Urdu, and Roman Urdu in real time, directly from their smartphones. By combining state-of-the-art NLP research with practical mobile application development, this study aims to bridge the gap between academic detection models and publicly accessible misinformation-fighting tools in a low-resource language and developing country context.

5. Objectives

1. To collect and curate a multilingual fake news dataset comprising verified real and fake news samples in English, Urdu, and Roman Urdu from Pakistani and international news sources, including Dawn, Geo News, ARY News, and social media platforms.
2. To develop and fine-tune a multilingual NLP classification model, using transformer architectures such as mBERT or XLM-RoBERTa, capable of detecting fake news and misinformation in English, Urdu, and Roman Urdu with high accuracy and F1-score.
3. To design and implement a hybrid deep learning pipeline combining CNN-LSTM and attention-based mechanisms, informed by the CNN-LSTM architecture of Hashmi et al. [2] and the DANN-based domain adaptation approach of Joshi et al. [6], to improve generalization across social media domains.
4. To integrate Explainable AI (XAI) techniques, specifically LIME, into the model pipeline to provide users with transparent, human-understandable explanations of why a piece of news has been classified as fake or credible.
5. To develop a cross-platform mobile application (Android/iOS) that allows users to input news text, URLs, or screenshots in any of the three supported languages, and receive a real-time credibility verdict along with a confidence score and explanation.
6. To evaluate the proposed system against existing tools (Google Fact Check Explorer, Snopes, ClaimBuster, Facterverse, Botometer) on a Pakistan-specific test set, showing measurable improvements in multilingual detection accuracy and user accessibility.

6. Methodology

The project will follow a systematic Research and Development methodology structured across five phases.

Phase 1 — Dataset Collection and Annotation: A multilingual fake news dataset will be compiled from Pakistani news portals (Dawn, Geo News, ARY News), international fact-checking repositories (IFCN-verified), and social media platforms (Twitter(X), Facebook, WhatsApp). The dataset will cover Formal English, formal Urdu (script), and Roman Urdu (transliterated). Manual annotation will be performed by trained annotators using a binary labelling scheme (Fake/Real), cross-validated against existing fact-check databases including Snopes and Google Fact Check Explorer. Data augmentation techniques like back-translation through Urdu as applied by Joshi et al. [6], will be used to address class imbalance.

Phase 2 — Model Development: A multilingual transformer model (mBERT or XLM-RoBERTa) will be fine-tuned on the selected and verified dataset. Additionally, a hybrid CNN-LSTM architecture with FastText multilingual embeddings will be developed and compared, inspired by Hashmi et al.'s CNN-LSTM approach [2]. The Domain Adversarial Neural Network (DANN) technique from Joshi et al. [6] will be adapted to enable generalization across different Pakistani social media domains (WhatsApp, Twitter, news portals). Hyperparameter tuning and regularization will be applied to prevent overfitting.

Phase 3 — Explainability Integration: LIME will be applied to the best-performing model to give explanations for each classification decision, which will be designed to be understandable to non-expert users and will be displayed in the mobile app along with the credibility verdict.

Phase 4 — Mobile Application Development: A cross-platform mobile application will be developed using React Native or Flutter. The application will provide three input modalities: manual text entry (supporting English, Urdu, and Roman Urdu), URL-based article analysis, and image/screenshot OCR-based text extraction. The backend model will be hosted as a REST API, with the mobile frontend consuming predictions in real time. UI/UX design will be guided by the findings of Lu et al. [4] regarding credibility indicator presentation timing.

Phase 5 — Evaluation and Testing: The complete system will be evaluated on a Pakistan-specific test set using standard metrics: Accuracy, Precision, Recall, and F1-Score. User acceptance testing will be conducted with a sample of Pakistani university students. Results will be compared against the reviewed tools to show the benefit and practical value of the proposed system.

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